Final Exam: Principles of Fixed Income

Cello Research

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Due: Monday, June 29th by 9:00am.

Problem 1. (10 points) You are considering the purchase of a home. It is in excellent condition, except for the roof. The roof has only 5 years of life remaining. A new roof lasts 20 years but costs $20,000. The house is expected to last forever. Assuming that costs will remain constant and the annual interest rate is 5% what value would you assign to the existing roof? [Hint: Consider the tradeoff in costs between replacing the roof now or waiting 5 years. Take into account the fact that a roof must be replaced every 20 years in your calculations.]

Problem 2. (10 points) Consider the two projects whose cash flows are shown in the table below. Find the IRRs of the two projects and the NPVs at an annual interest rate of 5%. Show that the IRR and NPV figures yield different recommendations. Which one would you rather pick? Justify your choice.

Years	Project 1 Cash Flow	Project 2 Cash Flow
0	-100	-150
1	30	42
2	30	42
3	30	42
4	30	42
5	30	42

Table 1: Cash flows for two projects

Problem 3. (10 points) An investor purchased a small apartment building for $250,000. She made a downpayment of $50,000 and financed the remaining balance with a 30-year fixed-rate mortgage at 12% annual interest, compounded monthly. For exactly 20 years she made equal-sized monthly payments as required by the terms of the loan. Now she has the opportunity to restructure the mortgage by refinancing the balance. She could borrow the current balance, pay off the original loan, and assume a new loan for this balance (Assume that no points or any charges are involved in the transaction). The new loan is a 20-year fixed-rate loan at 9% annual interest, compounded monthly, to be paid in equal monthly installments. Suppose that she receives an annual rate of 5% on her risk-free savings account. Should she restructure the mortgage?

Problem 4. (15 points) Assume a continuous compounding convention for this problem.

1. What is the (modified) duration of a 20-year zero-coupon bond?
2. What is the convexity of a 20-year zero-coupon bond?
3. A portfolio manager tells you: “Convexity is the change in duration as a function of interest rates and since the duration of a zero-coupon bond is a constant that is independent of the interest rate, it follows that the convexity of a zero-coupon bond must be zero.” Does this statement agree with your calculation above? If not, can you explain the flaw in the portfolio manager’s reasoning?

Problem 5. (15 points) Portfolio X is only invested in zero-coupon bonds of a fixed maturity. It lost \$1.6 billion out of a market value of \$7.5 billion in assets because of an unexpected increase in interest rates from 3% to 5.7%. Assume all interest rates are continously compounded.

1. What was the maturity of the zero-coupon bonds in Portfolio X?
2. By borrowing money, a new portfolio (Portfolio Y) of \$7.5 billion in zero-coupon bonds of a fixed maturity was levered up to \$20.5 billion. Assume that the liability was taken on through repo positions which are financed at the overnight rate. What is the total amount of the liability (L) necessary to lever up the portfolio and what is its duration (D_L)?
3. What maturity do the zero-coupon bonds in Portfolio Y have to have in order to generate the same losses as Portfolio X?
4. (**Extra Credit: 10 points**) A portfolio with both assets (A) and liabilities (L) has an equity (E) given by $E = A - L$. By treating equity as a portfolio, calculate the duration of equity D_E as a function of duration of the assets D_A and liabilities D_L. Use this formula to calculate the duration of the levered portfolio (Portfolio Y). What conclusions about the riskiness of Portfolio X versus Portfolio Y can you reach by going through this exercise?

Problem 6. (15 points) The principles of cash flow analysis can be used to evaluate the value of publicly traded corporations. The owner of a share of stock in a company can be expected to receive periodic dividends. Suppose that it is known that in year k, $k = 1, 2, \ldots$, a dividend of D_k will be received. If the interest rate is fixed at r, it is reasonable to assign a value of the firm to the stockholders at the present value of this infinite dividend stream; namely:

$$V_0 = \frac{D_1}{1+r} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \cdots$$

Note that the formula requires the future dividends be known. A popular way to specify dividends is to use the **constant-dividend model**, where dividends grow at a constant rate g. In particular, we assume that $D_{k+1} = (1+g)D_k$.

1. Under what assumptions does the infinite series expression for V_0 have a finite value?
2. Assuming that the summation has finite value, find a formula for V_0.
3. The XX Corporation has just paid a dividend of \$1.37mm. The company is expected to grow at 10% for the foreseeable future, and hence most analysts project a similar growth in dividends. The discount rate used for this type of company is 15%. What is the value of a share of stock in the XX Corporation? (Assume that there are 1 million shares outstanding.)

Problem 7. (45 points) Often a mortgage payment is divided into a principal payment stream and an interest payment stream, and the two streams are sold separately. We will calculate the values of these individual streams in this problem. Consider a standard fixed-rate level-payment mortgage with an initial balance $B = B(0)$ with equal periodic payments of an amount M. Assume that the interest rate per period is r.

1. Show that the mortgage balance $B(k)$ after the kth payment satisfies:

   $$B(k) = (1+r)B(k-1) - M$$

   for $k = 0, 1, \ldots,$.
2. Show that the equation above has the solution:

   $$B(k) = (1+r)^k B - \left[\frac{(1+r)^k - 1}{r}\right] M$$

3. Suppose that the mortgage has N periods and M is chosen so that $B(N) = 0$; show that:
$$M = \frac{r(1+r)^N B}{(1+r)^N - 1}$$
4. The kth payment has a principal component of $P(k) = M - rB(k-1)$. Show that $P(k) = (1+r)^{k-1}(M - rB)$.
5. Find the present value V (at rate r) of the principal payment stream in terms of B, r, n, M.
6. Find V in terms of r, n, M only.
7. What is the present value W of the interest payment stream $I(k) = rB(k-1)$ in terms of B and V?
8. What is the value of V as $N \to \infty$?
9. For a given N, which stream do you think has the larger duration – principal or interest?

Problem 8. (25 points) Assume that all interest rates are compounded annually for this question. Suppose that the prices of \$1,000 face zero-coupon bonds are as follows:

Maturity	Price
1 year	\$950
2 year	\$890
3 year	\$830
4 year	\$770
5 year	\$720

Table 2: Zero coupon prices

1. Calculate the value of an instrument paying \$1,500 at the end of Year 2 and \$2,000 at the end of Year 5 without computing spot rates.
2. Compute the spot rates of interest associated with these zero-coupon prices.
3. Compute the implicit forward rates $f_t = f_{t,t+1}$, where $f_{t,t+1}$ is the forward rate that applies between time t and $t+1$ for $t = 0, 1, 2, 3, 4$ years.
4. By using the general formula that connects the forward rate $f_{i,j}$ to the spot rates s_i and s_j, show that the t-year spot rate s_t is given by the formula for $t \geq 2$:
$$s_t = \sqrt[t]{(1+f_0)(1+f_1)\cdots(1+f_{t-1})} - 1$$
This formula shows that spot rates can be considered geometric averages of the forward rates of interest.
5. Using the forward rates $f_{t,t+1}$ computed previously, calculate the price of a zero-coupon bond that pays \$10,000 at the end of Year 4. Does the computed value match the price shown in the table? Why or why not?

Problem 9. (10 points) Suppose a Central Bank wants to achieve a steady inflation rate. Now suppose that it starts to see nominal interest rates rise in the financial market. How should it respond to this increase? Should it increase or decrease the money supply?

Problem 10. (10 points) Investors wonder how the Fed will react. The U.S. inflation rate is expected to be 1%, and the Congressional Budget Office estimates that current real GDP is 5% below potential GDP.

1. Use the Taylor rule to predict the federal funds target rate. Assume the Fed's inflation target is 2%, the long-run real interest rate is 2%, and the sensitivity parameters (or coefficients) for deviations of inflation from its target and for deviations of output from its potential are 1.5 and and -0.5 respectively.
2. Why can't the Fed implement this policy target? Be specific.